agents such as viruses, bacteria or toxins. So acquired immunity has memory. The innate immunity and acquired immunity work together to eliminate pathogenic microorganisms and other foreign agents. Although innate systems predominate immediately upon initial exposure to foreign substances, multiple bridges occur between innate and acquired immune system components.

13.4.2 Mechanisms of Innate Immunity

A potential microbial pathogen invading a human host immediately confronts a vast array of nonspecific defence mechanisms. Many direct factors (nutrition, physiology, fever, age, genetics) and equally as many indirect factors (personal hygiene, socioeconomic status, living conditions) influence all host microbe relationships. In addition to these direct and indirect factors, a vertebrate host has the following four non specific defence mechanisms.

A. Physical barriers
B. Chemical mediators
C. Phagocytosis
D. Inflammation

A. Physical barriers

i) Skin

Intact skin contributes greatly to host resistance. It forms a very effective mechanical barrier to microbial invasion. Its outer layer consists of thick, closely packed cells called keratinocytes, The skin is slightly acidic (around pH 5-6) due to skin oil, secretion from sweat glands and organic acids produced by commensal *Staphylococci*. It also contains a high concentration of sodium chloride and is subject to periodic drying.

ii) Mucous membranes

The mucous membranes of the eye (conjunctiva), the respiratory, digestive and urogenital systems withstand microbial invasion. One antibacterial substance in these secretions is lysozyme, an enzyme that lyses bacteria. Mucous secretions possess the iron binding protein, lactoferrin. Lactoferrin sequesters iron from the plasma reducing the amount of iron available to invading microbial pathogens and prevents their ability to multiply. Mucous membranes produce lactoperoxidase, an enzyme that catalyzes the production of superoxide radicals, reactive oxygen intermediate that is toxic to many microorganisms.

iii) Respiratory system

Microbes smaller than 10μm pass through the nasal cavity and are trapped by the mucociliary blanket and the trapped microbes are transported by ciliary action that moves them away from lungs. Coughing and sneezing reflexes clear the respiratory system of microorganisms by expelling air forcefully from the lungs through the mouth and nose, respectively. Salivation also washes microorganisms from the mouth and nasopharyngeal areas into the stomach.

iv) Gastrointestinal tract

Most microorganisms that reach the stomach are killed by gastric juice. (pH 2-3). However, organisms embedded in food particles are protected from gastric juice and reach the small intestine. The mucous membranes of the intestinal tract contain paneth cells. These cells produce lysozyme and cryptins (toxic for bacteria).

v) Genitourinary tract

Under normal circumstances, the kidneys, ureters and urinary bladder of mammals

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